QP code: **25901642**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE

EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCCSC100 - DATA VISUALIZATION USING PYTHON

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Scheme of Evaluation

Part A

Very Short Answer Questions -Answer all questions - Each question carries 2 marks

1. Find the function used to create a basic line plot in Matplotlib? [K] / [CO1]

Ans: **plot()** function connects the data points with a line, producing a simple line graph.

2. What is Seaborn, and why is it used for data visualization? [K] / [CO1]

Ans: **Seaborn** is a Python data visualization library built on top of Matplotlib.
It is used to create attractive and informative statistical graphs easily, with better design and simpler code.
1 mark for definition and 1 mark for use.

3. Name two types of input components available in Dash [K] / [CO2]

Ans: Dropdown (dcc.Dropdown) ,

Input box (dcc.Input) Any 2 types– 2marks

4. Name any 2 static plotting library [K] / [CO2]

Ans: Matplotlib, Seaborn

5. Can we save the charts as image files in Plotly? Describe how [U] / [CO2]

Ans: Yes, charts in **Plotly** can be saved as image files. – 1 Mark

This is done using the **write_image()** function after installing *kaleido*,. – 1 Mark

e.g. fig.write_image("chart.png").

Part B

Short Answer Questions -Answer any 5 questions- Each question carries 5 marks

6. Relate data visualization techniques to real-world decision-making processes. [U] / [CO1]

Ans: Data visualization techniques play a key role in real-world decision-making by

transforming complex data into clear, understandable visuals.

- **Better understanding of data:** Charts and graphs (bar charts, line graphs) help decision-makers quickly grasp trends and patterns.
- **Identifying relationships and trends:** Visuals like scatter plots and heatmaps reveal correlations and hidden insights.
- **Faster decision-making:** Visual representation reduces time needed to analyze large datasets.
- **Improved communication:** Visuals make it easier to present findings to stakeholders and support arguments.
- **Risk reduction:** By clearly showing patterns and anomalies, visualization helps in making informed and accurate decisions.

Thus, data visualization helps organizations and individuals make smarter, evidence-based decisions in areas like business, healthcare, education, and government.

7. Discuss different methods used in Plotly express for quick visualization with necessary codes. [U] / [CO2]

Ans: **1. Line Plot – px.line()**

Used to show trends over time. Write the program segements for each plot

2. Bar Chart – px.bar()

Used for comparing categories.

3. Scatter Plot – px.scatter()

Used to show relationships between variables.

4. Histogram – px.histogram()

Used to show data distribution.

5. Pie Chart – px.pie()

Used to show proportions.

These methods help in quick visualization with fewer lines of code and provide interactive features like zooming and hovering.

Each method with code → **1 mark × 4 methods = 4 marks**

Overall explanation → **1 mark**

8. What is a multi-plot grid in Seaborn? How is it useful? [K] / [CO1]

Ans: A **multi-plot grid in Seaborn** refers to a structured way of displaying multiple plots in a grid layout, where each subplot represents a subset of the data. Seaborn provides built-in classes like **FacetGrid**, **PairGrid**, and **JointGrid** to create such visualizations.

Uses:

- It helps in **comparing patterns across different subsets** of data easily.
- Makes it simple to **visualize relationships between variables** under different conditions.
- Improves **clarity and organization** instead of crowding all data into one plot.
- Useful for **detecting trends, variations, and outliers** across categories.
- Widely used in **exploratory data analysis (EDA)** for deeper insights.

Definition- 3 Marks , Uses- 2 Marks

9. Describe the role of storytelling in data visualization and how it improves comprehension?

[U] / [CO2]

Ans: Storytelling in data visualization refers to presenting data with a clear narrative so that it conveys meaningful insights instead of just raw numbers or charts.

How it improves comprehension

- Makes complex data **simple and easy to understand**.
- Helps in **quick identification of trends and patterns**.
- Improves **memory and engagement** compared to plain data.
- Supports **better decision-making** by clearly conveying insights.

Definition 2 Marks , Any three improvisations- 3 Marks

10. Explain the different types of charts and their appropriate use cases in data visualization. Use appropriate diagrams. [U] / [CO2]

Ans: Different charts are used based on the **type of data and purpose**:

- Comparison → Bar chart
- Trend → Line chart
- Proportion → Pie chart
- Distribution → Histogram/Box plot
- Relationship → Scatter plot

1Mark for each Chart with diagram and Explanations

11. Compare the effectiveness of line plots and scatter plots in data representation? [U] / [CO1]

Ans: **Line plots** and **scatter plots** are both used to show relationships between variables

- **Trend analysis:** Line plot is more effective.
- **Relationship/correlation:** Scatter plot is more effective.
- **Data continuity:** Line plot emphasizes continuity; scatter plot does not.
- **Outlier detection:** Scatter plot is better.

Line plots – 2.5 marks Scatter plots- 2.5marks

12. Differentiate Plotly Express and Plotly Grpah Objects [U] / [CO2]

Ans: **Plotly Express**

- A **high-level interface** for creating plots quickly.
- Requires **less code** and is easy to use.
- Built on top of Graph Objects.

- Automatically handles **data mapping, labels, and styling**.
- Suitable for **quick visualization and exploratory data analysis (EDA)**.

Plotly Graph Objects

- A **low-level, detailed interface**.
- Requires **more code and manual control**.
- Allows full customization of every element (layout, axes, annotations).
- Suitable for **complex and highly customized visualizations**.

Plotly Express -2.5 Marks Plotly Graph Objects -2.5 Marks